CS-104 Quiz 2

Fall 2026

Name: _____

This is an in-class, closed-book quiz (30 points). If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.

1. (5 points)

Assume `x = 4`, `y = 9`, and `name = 'Ada'`. Give the value of each Python expression below (½ point each). If it raises an error instead, name the error.

```
>>> not True
```
False — example answer

```
>>> x > 3 and y < 9
```

```
>>> x > 3 or y < 9
```

```
>>> not x == 4
```

```
>>> 3 < x <= 4
```

```
>>> x % 2 == 0 and y % 2 == 0
```

```
>>> 'd' in name
```

```
>>> 'A' in name.lower()
```

```
>>> y == 4 or 5
```

```
>>> not (x > y)
```

```
>>> x + y > 12 and name == 'ada'
```

2. (4 points)

What does this code print? Write every line of output.

```
total = 0
for i in range(1, 10, 3):
    if i % 2 == 0:
        total = total + i
    else:
        total = total - 1
    print(i, total)
print("done", total)
```

3. (7 points)

Write a code segment that computes a shipping cost. It inputs a package weight in kg (as a float) and whether the shipping is express (the user types `yes` or `no`). Then:

- If the weight is 0 or less, print `invalid weight` and nothing else.
- Otherwise, the base cost is $5 for up to 1 kg, $9 for more than 1 kg and up to 5 kg, and for more than 5 kg it is $9 plus $2 for every kg above 5 (e.g. 7.5 kg costs 9 + 2 × 2.5 = $14).
- Express shipping adds $10 to the base cost.
- Print the final cost, e.g. `Cost: 24.0`.

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

4. (2 points)

Fill in the blanks in each `range(...)` call so the loop prints exactly the numbers described (1 point each).

a) Every multiple of 7 from 7 to 70, inclusive:

```
for n in range(___, ___, ___):
    print(n)
```

b) The numbers 30, 27, 24, ..., 6, 3 (counting down by 3):

```
for n in range(___, ___, ___):
    print(n)
```

5. (4 points)

Consider the following (syntactically correct) code segment:

```
count = 0
for i in range(10, 0, -2):
    count = count + i
print(count)
```

a) What does it print? b) Write a `while` loop that does the same thing.

6. (8 points)

Write a code segment that repeatedly inputs whole numbers from the user until the user enters `0`. Then it prints how many numbers were entered (not counting the 0), their sum, and the largest one. Do **not** use the built-in `max` function or a list. If no numbers were entered before the 0, print `no numbers` instead. Example run:

```
Number: 4
Number: 11
Number: -2
Number: 0
Count: 3, sum: 13, largest: 11
```

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.